

# Cross-Project Bug Localization: Final Fix List

To A Eashaan Rao. From Sridhar Chimalakonda. RISHA Lab, IIT Tirupati. Reference: version 8 of the manuscript, 28 pages. Target venue: Journal of Systems and Software.

## Core Argument

Version 8 is close. The claim softening has largely landed, Figure 3 is now honest, the tie footnote reconciles, and the paper is two pages shorter. What remains falls into six groups. Sections A to C must be done. Section D is the cautionary reframe, which is reordering rather than rewriting and removes three overclaims as a side effect. Sections E and F are the residue. No new experiments are needed for any of it. Assuming the metric definitions resolve as Section B expects, this is roughly two days of work and then it should go out.

## A. Two broken citations (do this first, five minutes)

Section 2.1 and Section 3.5 both contain a bibliography entry compiling to a literal question mark:

using measures such as BM25 (?) or VSM over code tokens

This appears twice, once in the first three pages a reviewer reads. Nothing else on this list costs as much per character. Add the BM25 reference (Robertson and Zaragoza) and rebuild.

## B. Finish the metric definitions

Half done in version 8. Section 3.4 now correctly stores `ground_truth_files` as a list, and Section 3.7 gives proper multi-relevant-document formulas for MRR and MAP. Two definitions were left behind, and both are singular where the ground truth is not:

1. Section 3.7, Top-K Recall: "fraction of bug reports for which the ground-truth file appears within the top K positions". Undefined when a bug has several relevant files.
2. Table 3 caption, Recall@300: "the fraction of bug reports for which the ground-truth file is retrieved within the shared FAISS top-300 candidate pool". Same problem.

This is not pedantry, because the numbers still do not reconcile as written. TRANP-CNN's MRR exceeds what its own Top-K values permit in three of four rows of Table 6 (0.326 against a maximum of 0.262, 0.389 against 0.291, 0.380 against 0.288), and Case 2 reports 0.654, 0.760 and 0.800 on a jupyterlab target whose Recall@300 is 47.0%.

The likely resolution is now visible. If Top-K and Recall@300 use a strict criterion, meaning all ground-truth files must appear within the top K, while MRR uses the best-ranked one, then every violation dissolves and no number moves. Please check the evaluation script, and if that is what it does, state it explicitly in both places. If it is something else, stop and tell me before continuing, because that case may move numbers in Table 6, Table 8 and all three cases in Section 4.4.

## C. Venue declarations (one hour)

Still entirely absent, and the only items on this list that can cost us before a reviewer opens the paper.

1. Highlights, three to five, each within 85 characters.
2. CRediT author contribution statement.
3. Declaration of competing interest.
4. Formal funding statement, beyond the title-page footnote.
5. RISHA Lab AI assistance disclosure.

6. Complete the ACM SIGSOFT empirical standards checklist for the General and Data Science standards, cite Ralph and colleagues 2021, and include the checklist in the Zenodo package. Add a short paragraph in Section 3 naming which standards the study follows.
7. Name the COOBA and TRANP-CNN implementations explicitly in Data Availability. Given the re-implementation threat they are the most important artifact in the package.

## D. The cautionary reframe

The paper currently has two centres of gravity pulling apart. RQ1 says CPL works and Section 5.1 gives confident deployment advice; RQ3 says outcomes do not generalise and everything needs validating. The abstract closes on RQ3, the conclusion closes on the methodological point, and Section 5.1 reads as though RQ1 is the finding.

Make the cautionary framing primary throughout. It is the more defensible claim, the more novel one, and the one the title promises. The rule: the variation across architectures is the finding, and RQ1 is evidence inside it, not a parallel headline. Every number stays; what changes is order and grammatical subject. Ten locations follow. Section 5.2's opening is already correct and is the model for the rest.

### Sentence-level replacements

#### D1 Abstract, results sentence

*Current: CPL outperforms WP-small for 2 of 3 architectures: BLAZE with a large effect... while COOBA shows only marginal gains...*

*Replace with: CPL's benefit is not uniform across the three: it outperforms WP-small for BLAZE with a large effect (Cohen's  $d = 1.19$ , win rate 90.5%,  $p < 0.001$ ) and for TRANP-CNN with a medium effect ( $d = 0.59$ , win rate 68.3%), while COOBA shows only marginal gains ( $d = 0.27$ ) with a 31.7% negative transfer rate.*

#### D2 Summary of findings box

*Current: Cross-project transfer outperforms within-project limited training for 2 of 3 architectures (BLAZE, TRANP-CNN) but yields only marginal benefit for the third (COOBA)...*

*Replace with: CPL's benefit varies substantially across the three models tested on identical data: large for BLAZE, medium for TRANP-CNN, and marginal for COOBA, with negative transfer affecting 3.2%, 27.0% and 31.7% of pairs respectively. Section 4 reports the full per-research-question breakdown.*

The concrete numbers also fix a separate problem: this box currently defers everything to Section 4 and says almost nothing. The compression went one step too far here.

#### D3 Section 4.1.2, closing paragraph

*Current: Taken together, 2 of 3 benchmarked architectural paradigms show statistically significant CPL benefit with a medium-to-large effect size. This cross-architecture replication is the primary evidence that CPL is a viable strategy for data-scarce projects, rather than an artefact of any particular model's inductive bias.*

*Replace with: Taken together, the three models span a wide range of CPL benefit under identical conditions. Two of the three, BLAZE and TRANP-CNN, meet the criterion above; COOBA does not. That two architecturally distinct models both show benefit suggests CPL is not solely an artefact of one model's inductive bias, though with one model per paradigm this is not a replication in the methodological sense.*

This is the most exposed passage in the paper. The replacement also retires the "cross-architecture replication" overclaim.

#### D4 RQ1 summary box, opening line

*Current: Cross-project transfer with minimal target labels significantly outperforms within-project limited training for 2 of 3 benchmarked architectural paradigms.*

*Replace with: CPL benefit under label scarcity differs sharply across the three models tested, from a large effect to none surviving correction.*

#### **D5 Section 5.1, preamble**

*Current: The results support six concrete recommendations for practitioners considering CPL deployment.*

*Replace with: Because CPL benefit varied substantially across the three models tested, the recommendations below are stated per model rather than for CPL in general. The first applies to BLAZE, the fourth to TRANP-CNN, and the remainder across all three.*

#### **D6 Section 5.1, item 1 heading and opening**

*Current: CPL is a low-risk, high-reward strategy for BLAZE. If a project has few labeled bug reports and BLAZE is the tool available, cross-project transfer is beneficial in the large majority of deployments...*

*Replace with: For BLAZE, CPL showed consistent benefit with low loss exposure in this corpus. Of the three models tested, BLAZE showed the most favourable risk profile. If a project has few labeled bug reports and BLAZE is the tool available, cross-project transfer was beneficial in the large majority of deployments (Table 7), with losses that are rare and within the run-to-run noise floor (Section 4). This profile did not carry over to the other two models, so it should not be read as a property of CPL generally.*

The first sentence becomes the item heading; the rest is the body.

#### **D7 Section 5.3, opening**

*Current: It establishes that CPL benefit cannot be assumed to generalize across model architectures: a finding from one CPL model does not reliably predict outcomes from a different architecture on the same project pairs.*

*Replace with: It suggests that CPL benefit may not generalize across model architectures: a finding from one CPL model did not, in this corpus, reliably predict outcomes from a different architecture on the same project pairs.*

Also change "is a methodological contribution" to "is the methodological contribution of this study" in the preceding sentence.

#### **D8 Conclusion, first results paragraph**

*Current: Cross-project transfer outperforms within-project limited training for 2 of 3 architectures – BLAZE and TRANP-CNN (Section 4) – while COOBA shows only marginal benefit with near-zero cold-start performance.*

*Replace with: The clearest result is how much CPL benefit varied across the three models under identical conditions. Cross-project transfer outperformed within-project limited training for BLAZE and TRANP-CNN, while COOBA showed only marginal benefit with near-zero cold-start performance (Section 4).*

Then move the sentence beginning "And CPL outcomes vary substantially across architectures even on the same source-target pairs" out of the limits paragraph and up into this one, since it is now the frame rather than a limit. The limits paragraph then opens on negative transfer.

#### **D9 Conclusion, practitioner paragraph**

*Current: For practitioners, BLAZE can, in most cases, be applied immediately using a within-domain source project with zero target annotations...*

*Replace with: For practitioners, the practical implication follows from the variation rather than from any single result: the right CPL strategy depends on which model is being deployed. For BLAZE, a within-domain source project can in most cases be applied immediately with zero target annotations, and its within-domain CP-transfer win rate against WP-small climbs to 93.3% once a modest set of target labels is added. TRANP-CNN requires only 20% of target labels and then matches full within-project training. COOBA warrants empirical validation before adoption.*

#### **D10 Section 4.1.2, the paradigm sentence**

*Current: We classify an architectural paradigm as exhibiting practically beneficial CPL if...*

*Replace with: We classify a model as exhibiting practically beneficial CPL if...*

In the same paragraph, "2 of 3 benchmarked architectural paradigms" becomes "2 of the 3 models tested" throughout. The caveat added to Section 3.2 was written to retire exactly this language.

## **E. Contributions and the agent objection**

### **E1. Rewrite the agentic-LLM paragraph in Section 2.2.**

This is the sharpest reviewer objection the paper faces and the current paragraph does not survive it. It claims the CPL setting is "the complement" of the agent setting, meaning newly open-sourced, proprietary,

privacy-constrained and sparse-report repositories. But eleven of our thirteen projects come from SWE-bench, which are precisely the popular, well-instrumented, test-suite-bearing repositories the paragraph says agents are advantaged on. A reviewer will find this by comparing Section 2.2 against Table 3.

The strongest available answer is already in the paper but buried at the end of the paragraph: agents do repair, not ranking. Agentless and SWE-agent select and patch a small number of files, whereas this study measures fine-grained ranking across an entire codebase at Top-1, Top-5 and Top-10 with MRR and MAP. On SWE-bench, agent file-localization accuracy is an input to patching rather than the reported output, so no direct comparison is currently available. Lead with that.

Rebuild the paragraph in four moves. First, the task difference: ranking a full codebase against generating a patch for a shortlist. Second, the evaluation difference: no agent baseline exists at the metrics we report, so no comparison is possible from this study. Third, and this must be added, acknowledge that our corpus is drawn largely from SWE-bench and therefore does not itself demonstrate the complementary setting, so the privacy and cost arguments are motivations for studying CPL rather than findings of this study. Fourth, close on the point already there, that the mechanism finding may inform how agents prioritise code during localization.

Delete the four listed scenarios, since we evidence none of them, and delete "Agent-based approaches are therefore promising", which we did not test either. Add one threats sentence: a reviewer may reasonably ask how these models compare against LLM-agent localization, and we cannot answer that from this study.

## **E2. Demote C3 and promote the risk spectrum.**

C3, the structured pair-selection strategy, is a compute-budget necessity rather than a contribution, and we concede as much twice, in Section 3.3 and again in conclusion validity where we admit the sample over-represents DS-by-DS pairs. Listing it as a contribution invites a reviewer to attack it as a limitation dressed up. Move it into Section 3 as a design decision, and put the negative-transfer risk spectrum in the third slot instead: 3.2% for BLAZE against 27.0% for TRANSP-CNN against 31.7% for COOBA. That number is actionable, is not reported anywhere else in the CPL literature, and is fully defensible.

## **E3. Close the WP-small premise gap.**

Section 3.6 honestly reports the range as 22 to 267 labelled bugs, then asserts that even 267 is a small-data regime. That assertion is not supported, and the introduction's motivating case is a project with dozens of mappings. A reviewer will ask why the baseline is proportional when the premise is absolute. One sentence acknowledging that the 20% convention is a design choice rather than a model of the data-scarce regime closes it, particularly since the RQ2 finding, that gain correlates with how few labels the target has, is the same observation from the other side.

## **E4. Address the IR comparator gap directly.**

The new Section 3.5 subsection, Why Dense Retrieval Not Lexical IR, argues for BGE as the candidate-generation stage, which was never in dispute. The open question is why BM25 was not included as a zero-label comparator, since Section 2.1 identifies IR methods as requiring no training data and therefore as the real alternative for a project at zero labels. As written the section reads as answering the objection while sidestepping it. Add one sentence stating plainly that classical IR was not evaluated as a zero-label comparator, so the cold-start result is relative to WP-small rather than to the full set of options open to an unlabelled project.

## **E5. Move the cold-start confound forward.**

Section 5.4 correctly identifies that the shared FAISS stage is not neutral: BLAZE re-scores the same BGE signal that selected the pool, with a better aggregation, while the other two apply an independent scoring function to a pool chosen by a representation they never see. That concession sits twenty pages after the claim it qualifies. Add a sentence to Section 4.1.4 at the point the cold-start result is first stated, and read the finding as "a BGE-based reranker deployed without target labels performs comparably to WP-small" rather than as a claim about cross-project transfer specifically.

## **F. Specific errors and residue**

### **Introduced or missed in version 8**

1. Section 4.1.2 still says TRANP-CNN win rates "range from 54.0% (Top-1) to 71.4% (MAP)". Table 7's Top-10 is now 47.6%, below the stated minimum. The COOBA sentence was updated correctly when the tie counts were fixed; this one was missed.
2. Table 9 jupyterlab exclusion, still backwards. Section 4.2.3 says the domain effect "weakens further for TRANP-CNN". Within-domain 76.7% against the excluded cross-domain 36.8% is a 39.9 point gap, wider than the 28 points in the table. The exclusion strengthens TRANP-CNN's domain sensitivity. And the within-domain figures are still not recomputed, though Group A is DS-by-DS and jupyterlab is a DS project, so within-domain changes too.
3. Joint versus sequential, still contradicting in three places. Section 4.2.1 was fixed and now reads "the joint source-plus-target training signal". Section 5.2 still says TRANP-CNN "benefits from source pre-training" and "source pre-training provides a quality of initialization". Case 3 still says "cross-project initialization plus 20% target labels" and describes learning from jupyterlab's corpus and then calibrating to scikit-learn. Section 3.6 explicitly rules out that schedule. The Section 5.2 instance matters most, since the paper's Why rests on that argument.
4. Section 3.8 still renders as "the -divergence theory", with the H symbol missing.

### Overclaims surviving the sweep

Four are handled by D3, D7 and D10 above. These remain:

1. Section 3.2, "This diversity provides a robust basis". Robust is on our own banned list and has survived every version.
2. Section 4.1.2, "A medium effect means the advantage is real and reproducible", against a single seed and a reported run-to-run spread reaching 0.29 MRR.
3. Section 5.3, "an open research direction not addressed by existing proposals", unhedged.
4. Challenge Ch1, "No prior CPL study has enforced this standard", unhedged while the five parallel novelty claims elsewhere were correctly softened to "to our knowledge".
5. Section 5.1 item 3 still carries the COOBA source-quality recommendation without the hedge that 4.3.2 now applies to the same finding. The conjecture is flagged in 4.3.2; the recommendation built on it is not.

Search strings for the sweep: robust, establish, paradigm, replication, pre-training, initialization.

### Lower priority

1. Power claim still ">99% power at  $d = 0.5$ ,  $n = 63$ ". Approximately 98% is accurate.
2. Section 5.4 still says each target recurs "approximately five times" while Case 2 says jupyterlab has twelve runs. Give the actual per-target distribution.
3. Check Figure 8's in-panel rho annotations against the Section 4.3.2 values of minus 0.00, minus 0.296 and minus 0.208. The earlier version disagreed.
4. Mixed British and American spelling: labour, standardised, tokenised, prioritise, visualises and artefact sit alongside organized and analyzed. Pick one.
5. Early-access citation years: Hosseini cited as 2017 but TSE volume 45 is 2019, Huo as 2019 but TSE volume 47 is 2021, Zhu as 2022 but TKDE volume 35 is 2023. Verify the bge-code-v1 release year, cited as 2024.
6. Verify that Zhu and colleagues 2021, TroBo, makes the criticism of TRANP-CNN's baseline we attribute to it in Section 2.2. The trailing Chandramohan, Li and Reddy triple does not support the clause it follows.
7. Reference list: Gomes de Oliveira Neto is sorted under O.

## Length

28 pages against a 24 target. If you want the last four, the three RQ summary boxes, the three cases in Section 4.4, and the six recommendations in Section 5.1 are where they are. Each restates material the reader already has. This is optional, but a shorter paper reviews better: it gives a reviewer less surface to object to and fewer places for the paper to disagree with itself.

## Sequence

| Order | What                                                                                                                          | Time       |
|-------|-------------------------------------------------------------------------------------------------------------------------------|------------|
| 1     | Section A, the two broken citations                                                                                           | 5 minutes  |
| 2     | Section B, metric definitions, and report back before continuing if the script does something other than the strict criterion | half a day |
| 3     | Section C, venue declarations                                                                                                 | 1 hour     |
| 4     | Section D, the ten cautionary reframe replacements                                                                            | 45 minutes |
| 5     | Section E, contributions and the agent paragraph                                                                              | half a day |
| 6     | Section F, errors, sweep and residue                                                                                          | half a day |
| 7     | Send to Ch                                                                                                                    |            |

The one rule for this pass: nothing here needs a new experiment, and nothing here should add length. Sections D and F are subtractive by design. If a fix is making the paper longer, it is the wrong fix.

The study is sound and the protocol is a real contribution. What is left is the last of the distance between what we measured and what the paper says we showed.

Sridhar Chimalakonda

*The original ideas, analysis, and research synthesis in this report are the intellectual contribution of the authors. AI-assisted tools were used to enhance the clarity and presentation of the writing.*